

# Software Requirements Specification (SRS)

## Adaptive AI Orchestration & Optimization System

### 1. Functional Requirements

#### ***1.1 Query Processing***

- Accept user queries through REST APIs
- Validate input requests
- Route queries to orchestration pipeline

#### ***1.2 Complexity Analysis***

- Analyze query complexity
- Detect retrieval-heavy queries
- Classify queries into Low, Medium, and High complexity

#### ***1.3 Dynamic Model Routing***

- Dynamically select optimal model/pipeline
- Balance latency, cost, and response quality

#### ***1.4 RAG System***

- Retrieve relevant documents using vector search
- Inject retrieved context into prompts
- Generate context-aware responses

#### ***1.5 Metrics Tracking***

- Track latency
- Track model usage
- Track strategy usage
- Track quality score

#### ***1.6 Evaluation Engine***

- Evaluate generated responses
- Generate quality scores
- Compare execution strategies

#### ***1.7 Learning Engine***

- Update routing probabilities
- Improve future decision-making
- Store historical execution data

#### ***1.8 Dashboard***

- Display latency trends
- Display routing decisions

- Display model usage
- Display evaluation statistics

## 2. Non-Functional Requirements

### 2.1 Performance

- Maintain reasonable response latency
- Support concurrent query execution
- Provide near real-time dashboard updates

### 2.2 Scalability

- Support additional models
- Support new APIs
- Support larger datasets
- Support future orchestration strategies

### 2.3 Reliability

- Handle API failures gracefully
- Maintain execution logs
- Reject invalid inputs safely

### 2.4 Maintainability

- Modular architecture
- Clean code organization
- Independent components
- Proper documentation

### 2.5 Security

- API key authentication
- Input validation
- Rate limiting
- Audit logging

### 2.6 Usability

- Simple chatbot interface
- Easy-to-understand dashboard
- Clear API responses

## 3. Tech Stack

| Layer                | Technology    |
|----------------------|---------------|
| Programming Language | Python        |
| Backend Framework    | FastAPI       |
| Web Server           | Uvicorn       |
| Database             | MongoDB Atlas |

|                     |                       |
|---------------------|-----------------------|
| Frontend Dashboard  | Streamlit             |
| Vector Search       | FAISS                 |
| Embeddings          | sentence-transformers |
| Local LLM Runtime   | Ollama                |
| Models              | TinyLlama, Phi-3 Mini |
| Version Control     | Git & GitHub          |
| Deployment          | Render                |
| Future Optimization | ONNX, OpenVINO        |